

AJAX Complete Notes

1. What is AJAX?

AJAX (Asynchronous JavaScript and XML) is a technique for creating fast and dynamic web pages. It allows web pages to update asynchronously by exchanging small amounts of data with the server behind the scenes.

2. How does AJAX work?

AJAX works as follows:

1. User triggers an event.
2. JS sends a request using XMLHttpRequest or fetch().
3. Server processes it.
4. Response is received.
5. Page is updated without reload.

3. AJAX using XMLHttpRequest (Vanilla JS)

```
var xhr = new XMLHttpRequest();
xhr.open("GET", "https://jsonplaceholder.typicode.com/posts/1", true);
xhr.onload = function () {
    if (xhr.status === 200) {
        console.log(xhr.responseText);
    }
};
xhr.send();
```

4. AJAX using jQuery

```
$.ajax({
    url: '/api/data',
    type: 'GET',
    dataType: 'json',
    success: function(data) { console.log(data); },
    error: function(err) { console.error('Error:', err); }
});
```

5. AJAX using Fetch with async/await

```
async function fetchData() {
    try {
```

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```
    const res = await fetch('/api/data');  
    const data = await res.json();  
    console.log(data);  
  } catch (error) {  
    console.error('Error:', error);  
  }  
}  
fetchData();
```